

FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL/ELECTRONIC
ENGINEERING TECHNOLOGY

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LECTURE NOTE:

BY

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INTRODUCTION:

The word computer is a complex word derived from the word computing. It can be broken down into different sections and sub-sections and referred to by any different people from different fields of discipline for a different task at any given period of time.

DEFINITION OF COMPUTER

A Computer: Is an electronic device which is capable of accepting data as an input to store and process it to give out useful information as an output for further usage. It can also be defined as an electric device that operates with the help of an instruction as input (Programs) to do a specific task by accepting the instruction (program) as an input data to store it or process it to produce the required information as an output.

1.2 HISTORICAL DEVELOPMENT OF COMPUTERS

Scientists had carried out different inventions in the 19th century to develop an electronic device that can be manipulated to carry out some certain task in the field of computation which was later called computer.

The historical development of computer can not be complete without mentioning the legends behind the different development stages.

- i. **Abacus:** Used by early Egyptians and Babylonians in the 19th century. It is a calculating machine said to have been in existence for over (200) two hundred years ago. It is a mechanized pebble counter in which beads are strung on wire string of iron-rod held in frames. The beads are strung along the wires when counting subtracting or adding.
- ii. **John Napier:** A Scottish mathematician in 1617 who devised a calculating machine (a set of rods) used as multiplication aids. The rods were carved from bones which were often called Napier bones.
- iii. **Blaise Pascal:** A French mathematician who invented the time calculating, as at that time, he was a teenage (of about 19) nineteen years of age, at that time he was motivated by the desire to reduce the numerical labour involved in

his fathers work. It wasn't a commercial success because of the constant break down.

iv.

Joseph Jacquard: A textile manufacturer in the 18th century (1802-1803) derived the machined called "Jacquard loom". The machine was programmed by means of a special punched cards which stored information about the required pattern in a cloth. The punched cards were stringed together side by side in a long continuous strip. It allows the machine to move up and down on the position of the holes on the cards, though by making the machine to weave the cloth in a particular patterns.

v.

Chanel Babbage: An English man who made an attempt to design a computer from a hand calculator. He develop his different analytical engines incorporated many features which are still present in the modern day computers such as;

- a. Data and program fed into a device using a suitable medium.
- b. Storage facilities
- c. A mechanized unit for calculation

vi.

Herman Hollerith: A statistician with US census Bureu who designed a punched card based machined in 1890 and 1900, the machine was to work on collecting the data for previous census (1880's) despite the substantial increase in population (50million to about 63mill). At that time it was estimated that Holleriths system had accomplished in 1 year 7months what would have taken (100) one hundred clarks (seven years and eleven month (7years and 11months) to do (it's one forth of the time estimated from experience manually). He owned the computing, tabulating and calculating company which later become the international business machine (IBM) which is today one of the larges computer manufacturing company in the world.

1.3 CLASSIFICATION OF COMPUTER

Computers are classified base on their generation, data process, size and complexity and capability.

1.3.1 CLASSIFICATION OF COMPUTER BY GENERATION

Due to age of the technology, computers can be classified with respect to their features and characteristics as follows;

i. First generation computers (1946-1958)

These are early computers that are based on vacuum tubes technology and are characterized by;

- a. Very large size approximately (10) ten feet tall
- b. Low processing speed (micro-second)
- c. High failure rate
- d. High cost
- e. Generate alot of heat
- ii. Second generation computer (1959-1964)

These computers are more reliable than the first generation computers because it uses transistor technology instead of electron tubes of the first generation. These computers have the following characteristics;

- a. More reliable than the 1st generation
- b. Faster than the 1st generation because processors operates in milliseconds
- c. Small in size compare to 1st generation
- d. Generate less heat.
- iii. Third generation computer (1965-1970)

These computers were built using Integrated Circuit (IC) instead of transistor technology used in the 2nd generation and are characterized by;

- a. Increased speed compared to earlier ones
- b. More accurate and presized
- c. Much smaller in size

- d. Low price
- e. Witnessed rapid growth

iv. Fourth generation of computers; (1971-Date)

These computers were based on large scale integration (LSI) on integrated circuits (I.C's) it is characterized by;

- a. Improved input and output devices
- b. Many language developed to solved difference of language problems and many more problems
- c. Lower in cost
- d. Extremely smaller in size than the earlier generation computers
- e. Faster processing in nano seconds
- f. Lead to introduction of first micro-computer
- g. Introduction of micro-processors
- h. Rapid development and expansion in computers usage
- v. Fifth Generation of computer

What is considered to be the fifth generation computers are presently under development stage in advanced countries like Japan, America, etc. this generation of computers will have a different structure from the earlier generation of computers. It's greatest characteristic is its ability to possessed intelligence and think like human.

1.4 CLASSIFICATION BY TYPE OF DATA PROCESSED

- a. **Digital computers:** These type of computers operates on discrete signals which can be in one of two input state at anytime. Discrete value occur in earth step of it's operation example; counting on one's fingers, digital watch with ting computer in it, mostly used for commercial application.

- b. **Analogue computer:** These type of computers operate on continues signal, which flow in wave forms such as with Ac electricity, temperature, etc. the output is in the form of smooth graph form which useful information can be read such as thermometer reading.

c. **Hybrid computer:** These are mostly used in technical operations and has an in-built modem, it is characterized with both analogue and digital combined in it.

1.5 CLASSIFICATION BY SIZE AND COMPLEXITY OR CAPABILITY

a. **Super computers:** Super computers are special computers designed for a particular purpose.

b. **Mainframe computer:** Characterized by large size, very powerful processing power and speed and support multi usage. (multiple uses)

c. **Mini computers:** A scale down of the mainframe computer, they are smaller in size compare to mainframe and easily manageable in course of operation and maintenance.

d. **Micro computer:** These are the smallest in size of all sizes of computers, with lesser speed and capability. They are based on the micro-processor used in producing them.

1.6 APPLICATION USED OF COMPUTERS

Computer uses and application in task in the modern world today is very wide and diverse that it cannot be exhausted below are few among the many uses and application of computers in our daily under takings.

- i. Used in our offices for word processing and desktop published etc.
- ii. Use the schools for record keeping and other administration works
- iii. Business applications in banks and super market for payrolls and other records
- iv. Used in industries for data processing
- v. Used in the field of engineering for engineering design and analysis
- vi. Artificial intelligence e.g robotics
- vii. At homes for games and entertainment

1.7 ADVANTAGE OF COMPUTERS OVER HUMAN BEINGS

- i. High speed which is comparable to lightning
- ii. Versatility with respect to input it can accept is very wide
- iii. It is very precise and accurate

- iv. It is reliable
- v. Has very large storage capacity
- vi. It is diligent

Though it has many capability but lacks one thing, it make no decision of it own, also it cannot think on its own and execute but work on the input received, therefore work on artificial intelegent, it can only work on the input data given to it as input (program) which it manipulate (i.e process) and give it out as an output information (result)

(Input) Data – Process – Information (output)

1.8 COMPUTER HARDWARE

The term computer hardware refers to the physical parts (components) of the computer system which can be seen, touched and flit by individuals.

1.9 FUNDAMENTAL COMPONENTS OF A COMPUTER

A computer system has four (4) fundamental components which includes;

- i. Input unit,
- ii. Central processing unit (CPU)
- iii. Storage unit
- iv. Output unit

Input Unit: This unit consist of all the devices used in entering data (usually unprocessed) into the computer system, they act as a get way of giving input into the computer usually in a user language form (understandable language from the outside world which the computer itself can understand example of input devices are; Keyboard, Mouse, Scanner, Lightpen, Microphone, Optical Mark, Recognition, Smart card etc.

Output Unit: This unit consist of all the device the computer used in transmitting information (result) to the outside world. The device transform the processed information in the computer which is normally in machine language to human understandable language, it work as an interface between the computer and its users.

Example of output devices includes, video display unit (VDU), printer, scanner, projector, speaker, e.t.c

(iii) **Central Processing Unit (CPU)**, in any computer system, this unit serves as the main brain and centre of the computers activities. The CPU represents the processing capability of the computer which is unique to computers. Because generally a computer system is recognized by the type and nature of its c p u. the CPU. is responsible for;

- (i) Co-ordinating the activities of all units or devices associated with the computer system.
- (ii) Accepting and executing instruction to cause the computer system to perform task and c p u. consist of these major parts;
 - Arithmetic and logic unit (ALU)
 - Control unit (CU)
 - Register.

ARITHMATIC AND LOGIC UNIT: is a digital circuit that performs arithmetic and logical operations. It is the fundamental building block of the CPU of a computer.

CONTROL UNIT: this unit co-ordinates the input and output devices of computer system it fetch out the codes of all instruction in the micro programs.

Register: The faster of all forms of computer data storage located inside the processor, the CPU access it directly.

iv. **Storage Unit:** This unit consist of storage devices and memories used to hold information in machine understandable form. The serve as store to keep information and instructions that are processed by the computer.

The storage unit or memory is where the computer stores information, it is classified into two (2) major types namely; the primary and secondary memory.

- i. **Primary Memory:** This is where jobs or data are stored temporarily inside the computer system during or before processing.

- ii. **Secondary Memory:** This is where an already processed data or partially processed data are stored in the computer system.

1.10 EXAMPLES OF PRIMARY AND SECONDARY MEMORY

Primary memory	Secondary memory
i. RAM (Random Access Memory)	Magnetic tape; magnetic disk
ii. ROM (Read Only Memory)	Floppy disk hard disk etc

1.10.1 RAM (Random Access Memory: This is the storage space used by the computer when it is executing instructions and processing data. It can be read, already and even written on. It is a volatile memory because it's contents are lost when there is power failure, it hold information temporarily.

1.10.2 ROM (Read Only Memory): The contents of the ROM can only be read but can not be altered of whatever information (data is on it it's unlike the RAM which you can write on, the ROM is non volatile memory, it store information permanently even where there is power failure the content still remain intact.

1.11 Other Storage Device: Fixed and removable disk are another storage device used in the computer system for storage, the fixed disk as the name implies are fixed in the computer but can be removed when servicing the computer system, the removable disk can be removed after the operation you want to carry out with it on the computer is finished example of such disc includes the floppy disc, external hard disc etc.

1.12 Computer Software: Software is the general name for set of program and data or instruction which direct the system hardware on what is required to be done at a particular time for a specified purpose.

Definition: Computer software is a collection of computer programs and related data that provide the instruction telling the computer what to do and how to do it.

1.13 TYPES OF SOFTWARE

Computer software are of two (2) major types which includes;

- i. Application software

ii. System software

Application software: These are programs that are designed to enable the computer to carry out human-oriented tasks; it is to enable the computer to be used or be applied to a specific human task for example spread sheet, banking, word processing, graphic application.

System Software (Operating system): This is the collection of instructions put together by the manufacturer, that when installed, the computer performs its designated function effectively and also helps in co-ordinating the operating of other software used in the system.

They are programs designed to carry out a task which are machine-oriented e.g. Ms, DOS, Windows 95, window 7, window 8, window 10 etc.

Programs: A program is an independent set of instructions which work to direct other computer hardware's to perform some certain specific task.

Programming: Is the process of designing and writing a computer program that allows the computer hardware to interface with the programmer and the system, leading to execution.

Programmer: Is an individual that writes/creates computer software or application by giving the computer specific programming instructions.

1.15 TITLE BARS AND MENU BARS

These are examples of computer controls that you operate using mouse or keyboard. The control helps in selecting command, change setting or work with windows, most programs contain dozens or even hundreds of commands (actions) you used to work the program. Many of the commands are organized under menus to keep the screen uncluttered, menus are hidden until you click their title in the menu bar located just below the title bar as can be seen in the diagram of fig 1.15 below.

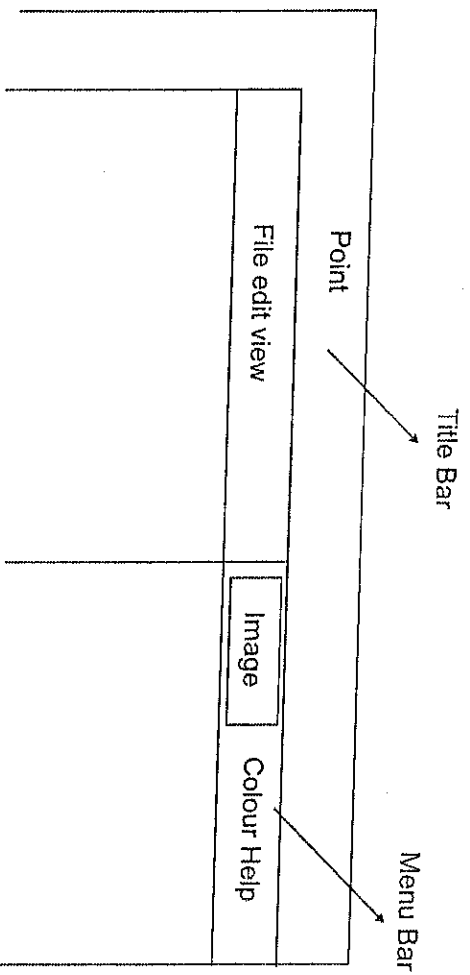


Fig 1.15 Diagram of a window

Point is the title of the entire program, image is title, it has flip/rotate, resize etc as the image menu.

1.16 Scroll Bar: When a document webpage or a picture exceeds the size of it's window, the scroll bar appear to allow you see the information that is currently not viewed fig. 1.16 shows an example of a scroll bar.

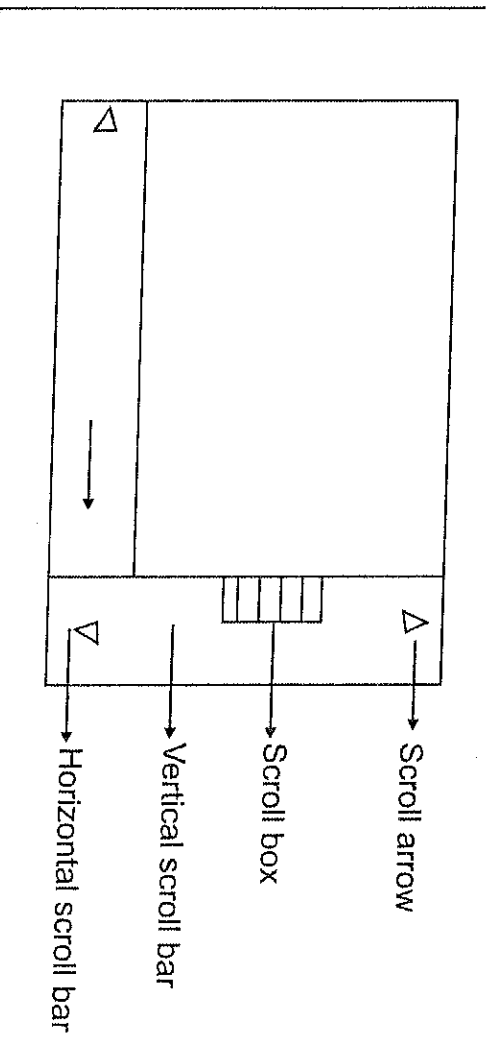


Fig. 1.16 Diagram of window showing scroll bar

To operate scroll bar, use any of the following.

- i. Click the up or down scroll arrows to scroll the windows content up or down in small steps hold down the mouse button to scroll continuously.

ii. Click on an empty area on the scroll bar above or below the scroll bar to scroll up or down one page.

iii. Drag the scroll bar up, down, left or right to scroll the window in that direction.

iv. If your mouse has a scroll wheel you can use it to scroll through document or web page, to scroll down roll the wheel back word and forward to scroll up.

1.17 Opening file and folder, cut, copying and moving files and folders

It is great important to open a file and folder menus before we can either cut, open, copy or move it.

1.17.1 File: A file is very much like a typed document that you might find on one's desk or in a filing cabinet. Its an item that contains a collection of related information examples; digital picture, songs, movies etc.

1.17.2 Folder: A folder provides a means of organizing and presenting your PC's resources such as files, programs etc, it helps to arrange files into logical groups. It can also hold other small folders called sub folders, examples of folder includes; my computer, my documents, my pictures my music folder etc.

1.18 OPENING A FILE OR FOLDER

The following steps are followed to open a file or folder;

- i. Double click on the file or folder you want to open and it would be displayed in a window.
- ii. Click on the start menu and click the file or folder once and it would open.
- iii. Right click on the file folder from the pop up window and click open then the file folder would open.

1.19 CREATING AND DELETING A FILE FOLDER

Some programs create a file when you open them, such programs as the word pads which start with a blank page (an empty unsaved file) start typing and when you are ready to do your work, click file in your menu bar and then click "save as" then type the file name which will help you locate the file again when needed in the future for use and click save.

When you no longer need a file, you can remove it from your computer, to delete open the folder, select the file and click delete, then select yes in the delete file dialog box, you can also delete by dragging a file to the recycle bin.

In creating a folder, you can also right click on the desktop choose the "new menu" then choose "folder", type your own description to the folder where new folder is written and then press enter.

1.20 CUT, COPY AND MOVE FILES AND FOLDER

To move files to a different folder, start by opening the folder that contain the file to be moved, then open the new folder where you what to move the file to, position the folder window on the desktop so you can see the content of both folders, next is to drag the folder or file (drag and drop method) to copy, right drag the file or folder from the first folder to the second folder.

Or

Right click on the file or folder you want to copy, cut, or move from the pop-up menu, select move, copy or cut. Open the location you what the item to be placed. Right click the location and selection paste, to delete you can also select the file or folder you wish to delete and from your key board press the delete key.

OPERATING SYSTEM

2.0 MICROSOFT WINDOW

2.1 Introduction: This topic introduces you to windows and show you how to do must of the common tasks. It provides you with tips and points out some exciting new features that come with windows.

2.2 WINDOWS AN OPERATING

Microsoft windows is an operating system, this means that it performs the function of an operating system. It is define as a system software that provide a graphical environment through which the reader can interact with the computer. The operating system uses windows to display information hence we call it window operating system. Windows can be moved and resized to display information.

2.2.1 VERSION OF WINDOWS

Windows has the following versions;

Windows 3.0, Windows 98, Window 7, Window 3.1, Windows me, ubuntu
Windows 95, Windows zero, Window 8, Window Xp etc.

2.2.2 FEATURES OF WINDOWS WAY TO START BROWSING

1. Windows explorer an easy way to start browsing through a managing network connectors, drives and files.
2. Long file name' up to 225 characters
3. New improved interface e.g task bar
4. The Microsoft network e.g enable you to watch movies on line and chat with friends.
5. Windows automatically set up your printer and hardware.
6. Multitasking i.e work with different programs
7. Enhance multimedia support and Games

2.3 STARTING YOUR COMPUTER

Before powering up your Pc make sure all the necessary cables are properly connected. Also it is important you ensure that the floppy disk drive no diskette. The reason for this is that as soon as the computer is on the system starts checking all attached devices such as; monitor, mouse, keyboard, printer etc. to ensure they are in good order.

The computer undergoes through a process called "Bootling" as the attached devices are being checked.

Definition: Bootling is the process of the computer system checking all the component of a computer system to ensure that all are ready to do work at one. It is also a process of starting/resetting the computer.

2.4 TYPES OF BOOTING

- i. Cold booting
- ii. Warm booting

Cold booting: When you turn the computer off and back on, your performing a cold boot, it enable all process to close and program update and installation to finish and can clear error.

Warm booting: When you restart the system without interrupting power it's a warm boot, it enable all process to close and program update and installation to finish and can clear error.

2.4.1 DEFINITION OF SOME WINDOW ACRONYMS

- i. **Windows desktop:** This is the background space on the screen where windows and icon are displayed after booting.
- ii. **Icon:** An icon is a graphical pictorial representation of files/folder command or a web page.

2.5 MICROSOFT DISK OPERATING SYSTEM (MS.DOS)

Introduction: MS-DOS is an acronym for Microsoft disk operating system, it is an old operating system that is used for managing, directing and controlling files that have been created with the application software. It provides a series of commands that help the system to communicate with the personal computers (PC).

2.5.1 Operating system: From fundamental knowledge on appreciation, we know that computer has two major elements the software and hardware.

Definition: Operating system is a set of packages that manages, controls, directs, and co-ordinates the whole activities of the computer system. E.g. unme, Ms 04. Again it can be seen as middle director of software and hardware by ensuring good relationship between both parties.

2.5.2 REASONS FOR STUDYING DOS

- i. It provide facilities for manipulation of the entire system.
- ii. It provides aids in manipulation which other packages cannot provides
- iii. It allow complete manipulation on store document or data

2.5.3 Starting up Dos: Before any system can be used, the operating system needs to be added for proper running of the computer system below are the two ways of loading Dos into the memory.

- i. Insert the Dos start-up disk into your computer and turn on the system. If your computer has a hard disk. Dos is automatically loaded whenever you turn on the system.
- ii. To start MS. Dos from windows, take the following steps (a) click the start button (b) select program, (c) Ms Dos prompt this take you to MS Dos environment.

Linux operating system

Introduction: Linux operating system is open source software that manages computer hardware resource and provide common services for computer programs. The operating is an essential components of the system software in the computer system. Linus only read from the disk those parts of a program that are actually used it consist of free and open source software for laptops, desk tops and servers it is a community developed. Commercially supported and offers free security update for at least 18 month often release with regular release every (6) six months it has a very clear focus on the user ability free security in the sense of given you right software freedom i.e the freedom to run, copy, distribute, study, share change and improve the software for any purpose without paying any licensing fee.

Fast and easy to install, installer ask a minimal of questions and the default will work for most people.

REQUIREMENT OF UNIX/LINUX

This has to do with the system RAM processor speed, compactability etc below is some requirement of the unix and linux operating system

- i. CD Rom Drive (for installation)
- ii. Net scape navigate 4.0 and above or Microsoft and internet explorer 4.0 and above is required.

- iii. Adobe acrobat reader 3.0 or above is required to view and print the MATLAB online documentation win in PDF format.
- iv. Some incense type requires a licence server running the MATLAB online document in document.
- v. TCP/IP is required on all platforms using lincees that requires a hardware lock.
- vi. Because of problem with the Java 1m, there four MATLAB on the HP, and IBM platform

MAIN FEATURES OF UNIX OS

- i. **Multuser:** More than one use can user the machine at the same time via terminal (serial or network connection)
- ii. **Multitasking:** More than one program can be run at a time.
- iii. **Multi platform:** Can be run on many CPU (Central Processing unit) not just intel.
- iv. **Multi proccession/Multithreading:** Is has native kernel support from multiple independent threads of control within a single process memory space, so that one program can not bring the whole system down.
- v. **Directory structure:** To support the organization and maintenance of file.
- vi. **Run on protected mode on the 386**
- vii. **Share copy or white pages among executable**
- viii. **Support for many national or customized key boards.**
- ix. **Virtual memory using paying (not swapping whole processes) to disk**

ADVANTAGES OF LINUX OPERATING SYSTEM

- i. **Security:** The security aspect of Linux is much stronger than that of windows need not to spend your money for antivirus software, the linux operating system has been in existence since early (90's) and has manage to stay secured in the realn of widespread for all time.

- ii. **Cost free:** Most obvious advantage of Linux operating system is the fact that it is free to obtain, while Microsoft product are available for heavily and sometimes recurring fee, Microsoft units typically are allowed to be installed on a single computer whereas linux distribution can be installed on any number of computers with out paying a single dime.
- iii. **Choice (freedom):** The power of choice is one of the great advantage of linux O.S. you have the power to control just about every aspect of the operating system (O.S).
- iv. **Software:** There are so many software of choice when it comes to distribution of specific task you could search for a text editor on fresh meat. and yield a hundred if not thousand.
- v. **Hardware:** Linux is perfect for those old computers with barely any processing power or memory, you have sitting in your garage basement collecting dust. Install linux and use it as a file server or a back-up server.

DISADVANTAGES OF LINUX O.S

- i. **Understanding:** Becoming familiar with the linux O.S requires patience as well as a strong learning curve you must have the desire to read and figure out things on your own rather than having everything done for you.
- ii. **Alternative program:** Through linux developers have done a great job at creating alternative to other popular windows applications, there are still some application that exist on window that have no equivalent linux applications.
- iii. **Small selection of peripheral hardware:** There is smaller selection of peripheral hardware drives for printers canners and other devices, it lack compactability with other hardware.

2.6 MS WORD (MICROSOFT WORD)

Word processing using MS. Word: A word processing package is a program, (i.e a standard package) which is written for typing editing processing, formatting serving

and printing of text, common examples are word perfect and Microsoft word (Ms-word)

Our choice for this lectures is MS word because with MS word you can create, letters proposals, business plans etc.

2.6.1 MS Word: This is a word processing package produced by Microsoft cooperation as an application software which is run through windows for letters, news letters, with pictures and fancy text effects, it has the following features:

- i. You can type, modify and store text for latter use
- ii. You can check spellings using the inbuilt dictionary which also suggest corrections for words wrongly spelt.
- iii. You can print your documents using attractive type faces

2.6.2 TO LOAD YOUR MS WORD

- i. Click on start button go to programs, click on "MS office" and click on your MS "Word"
- ii. Click start button choose new office document click on blank document to load "MS Word".

2.6.3 ENTERING A TEXT

When you load your MS word, a blank page comes up on the screen thereby giving you the right to type what you want. Or

Open the file menu and select new select blank document click the ok button start typing.

2.6.4 FORMATTING WITH WORD

When you want to emphasize text to make a point you can underline bold face or italicize your text.

- To underline, select the text you want to change and click "U" on the toolbar menu.
- To make bold, select the text you want to change and click "B" on the tool bar menu

- To use italics select the text you want to change and click I on the tool bar menu.

Or

You can also click the format Menu, click on font and choose whatever you want from the available options.

2.6.5 SAVING YOUR DOCUMENT STEPS

- Click the file menu
- Select “save as” for first time (If the file exist click “save”). But if you don't want your document to read default name as “my document” set the name you want the document to bear and click save.

2.6.6 PARAGRAPH ALIGNMENT

- Step 1: Click any where in the paragraph you want to change
- Step 2: Click the format menu and then click paragraph under alignment. Click the alignment you want either left, right or center depending on your choice of the paragraph alignment. To indent repeat step 1 and 2 above and under indentation type the measurement for how far you want to indent the paragraph.

2.6.7 IMPORTING OBJECT

Drag the mouse to the file you want to copy from click and highlight what you want to copy, click on the edit men and then click copy to copy or click control C to copy the selected information to the clipboard. Go to the file or folder you want to put the information you copied to the clipboard, click the edit menu and click paste then the copied information world now be inserted or click control V, i.e CTRL + V) to paste and CTRL C to copy while CTRL A to highlight.

2.6.8 EDITING TEXT

To edit any information move the mouse cursor to the editing area and click then start to edit what you want to edit, if there are other you want to edit scroll to the part and click the cursor to edit, either correction addition or to clean what you don't want.

2.6.9 GRAMMER AND SPELLING ~~CHECKING~~ ^{Checking}

Anytime you are typing and you see a word under lined with red or grey colour, MS. Word is letting you know that either the word is spelled wrongly or the grammar is wrongly arranged such can be corrected by:

- i. Right click on the misspelled word or line and a pop-up menu will be displayed which allow you to select the correct word or grammar or select ignore all if you are sure of what you write.
- ii. Choose spelling option pop-up menu to display spelling dialog box.
- iii. Choose the right suggestion of the misspelled word and click change all to change all the wrong spelled word to the correct spelling of the selected word.

TO CORRECT SUSPICIOUS GRAMMER FOLLOW

- i. Select spelling and grammar option from the tool menu replace the underline grammar with the suggested correct one.

2.7.0 USING THE SAURUS

Saurus produces a list of synonyms (i.e word with similar meaning).

- i. Select a word
- ii. Click on toll menu and choose language
- iii. Click on the saurus (it displays the various meaning to the word selected.
- iv. Click on the word you wish to replace (selected one)
- v. If you still want additional synonymous, click on look-up for MS. Word to look-up for more additional ones.

2.7.1 PRINTING OF DOCUMENT

- i. Select file from the menu bar.
- ii. Click on the print option
- iii. On the dialog box print number of copies you want to print
- iv. Set page to print
- v. Select print then ok

NOTE:

The print dialog box gives you many options/settings regarding to what you intend to print and you can adjust the option to meet your specification for print need such as paper size, page set up, colour, number of copies quality of print, pattern of print (Landscape, portrait letter).

2.7.2 INSERTING TABLE

Tables are a great way to show information, an easy way to read and useful statistic or list of items can be compared side by side.

To insert tables you must;

- i. Click on table menu and click insert
- ii. Click on the columns box and type the number of columns required.
- iii. Enter the number of rows needed
- iv. Click Ok

To draw a table

- i. Click on the “Table” menu and choose “draw table:..

If the mouse pointer turns to the shape of a pencil, click where you want to draw the table and click to start drawing drag the pointer diagonally up or down to draws the table as you would with a pencil.

To create enough rows or columns for your table or to delete rows or columns from your table follow;

- i. Select multiple rows or columns and right click your mouse to display a pop-up menu.
- ii. Choose insert rows or columns as the case may be and specify the number of rows or columns you want click otherwise choose delete rows or columns to delete.

2.7.3 SOME TERMS USED IN MS. WORD

- i. **Indentation:** Is a space between paragraphs edge and the page margin.

- ii. **Cutter:** Is the left side of a page where you might punch hole or attach a border.
- iii. **A window:** Is a paragraphs last line that print at the bottom of a page.
- iv. **A hand indent:** Is an indentation of all paragraph except the first line.
- v. **An orphan:** Is a paragraphs first line that print at the bottom of a page
- vi. **A font:** Any character style that determine the look and size of your character.
- vii. **The Saurus:** List of words of a similar meaning to each other. In other words the saurus produces a list of synonyms.

3.0 MS EXCEL

Spread sheet package using excel: Excel is a spread package that you can use to enter, manipulate and analyse data to your hearts desire. It helps in managing and presenting a data.

3.1 FUNCTIONS OF EXCEL

It is used in presentation of,

*M.S. Muranthu
O.P. operating System*

- i. Result sheet
- ii. Statistics
- iii. Pay roll
- iv. Used in saving equation in mathematics
- v. Also used in consolidation of account etc

3.2 WHAT YOU NEED TO KNOW BEFORE USING EXCEL

- i. **Worksheet:** It is the application workspace of a spread sheet program where you enter and manipulate data, it is made-up of cells organized into columns and rows.
- ii. **Cell:** It is the space at the intersection of rows and columns in a worksheet. It represent the basic unit of data entry in that each independent piece of data must be entered into a cell.

- iii. **A cell address:** It is a unique reference that identify each cell in a worksheet, its is formed from the combination of the columns letters and row numbers that form the cell e.g A1, B1, C1.....
- iv. **Cell pointer:** It is an indicator that identify an active cell.
- v. **Active cell:** It is the current cell that will be affected by data entry and manipulation operations.
- vi. **Formular Bar:** Is where you can enter and edit data related to formular.
- vii. **Workbook:** A Microsoft excel file is called a workbook, graph and other associated object each sheet name appears on a tap at the bottom of the workbook you can switch between sheets by a click on the sheet tabs at the bottom of the work book.

3.3.3 LOADING MS. EXCEL

- i. Click on the start button in the task bar
- ii. Click on the Microsoft office
- iii. Click on the Microsoft office excel its will open and the page that appears is called a worksheet.

3.3.1 ENTERING DATA INTO A WORKSHEET

- i. Position the cell pointer on the cell into which you want to type data (it become bold)
- ii. Type the data into the cell
- iii. Press the “enter key” or any of the movement keys on the keyboard.

3.3.2 SELECTING A GROUP OF CELLS ON A WORKSHEET

- i. Click on the first cell, hold and drag your mouse or hold down your shift key as you click on the last cell.
- ii. If its non-adjecent cell, press the ctrl (control) key and click on the other cells of your choice.
- iii. To select an entire columns, click the column heading

- iv. To select and entire row, click the row heading

Click any area outside the selected text location to cancel any selection.

3.3.3 EDITING

To edit any cell, double click on the cell with the cursor blinking, you then edit and presenter when you are through or use the arrow keys to move out or f2 in the cell to edit.

3.3.4 MOVING AROUND THE SHEET

You can move to the beginning of the sheet by pressing ctrl + home or to move to the last cell of the sheet by presenting ctrl + end from your keyboard or use your mouse to move the cursor and click where you want it to be.

3.3.5 FINDING AND REPLACING

- i. Click on "Edit" from the menu bar
- ii. Click on find (specify what you want)
- iii. Click on replace button (specifying what you want to replace)
- iv. Click on either (replace or replace M)

3.3.6 SAVING

If new;

- i. Click "file" on the menu bar
- ii. Click "save as" and type the name you what the file to bear
- iii. Click Ok

If not new

- i. Click file on the menu bar
- ii. Click "save"
- iii. Click Ok

3.3.7 FORMATING DATA

The steps below shows the specification of how data is displayed in the cell.

- i. Click on the format option on the menu bar
- ii. Click on style

- iii. Click on modify to adjust what you intend to adjust such as decimal place, alphanumeric, alignment etc changing column width row height, font style size etc.

3.3.8 COPY, MOVE/CUT, DELETE

- i. Highlight the text you wish to work on
- ii. Click edit from the menu bar
- iii. Click copy, delete, move or cut to perform any of the function you selected and for copy, move and cut open where you want it to be.
- iv. Click edit from the menu bar
- v. Click paste Or

Highlight the text and press delete for delete or ctrl+c for copy or ctrl+v for paste etc.

3.3.9 INSERTING AND DELETING SHEETS

Click on insert on the menu bar

Click worksheet and click insert

Click delete a sheet to delete

To Start a new workbook

Click on file on the menu bar

Click new

Click Ok

However to close the workbook click on file on the menu bar and click close.

3.4 CREATING FORMULAR IN CELLS

Important things to note while creating a formular in cells is that

- Note the cell address that contain the data you wish to work on.
- Place your cell indicator on the cell that will contain the result of your calculator.
- Assuring you want to sum cells B7, B8 type = sum (B7 +B8) and press enter

Note the following

- i. Select range of the data you want
- ii. Click on insert on the menu bar
- iii. Choose chart (A dialogue box comes up)
- iv. Choose the type of chart you want from the box
- v. Click next button (The chart source dialogue box appear
- vi. Specify/modify your chart data range and your series as desired (row and columns)
- vii. Click on next
 - a. Then you can add titles to your chart or graph
 - b. Suppress any of the exist (X or Y)
 - c. Add or suppress grid lines
 - d. Put data cables on the data points etc
- viii. Click the next button

You can now specify where you want your chart to be displayed, i.e as part of the current sheet or on a new separate sheet.

- ix. Click finish button

To edit a chart right click on any open space inside the chart and choose chart type edit as required, to move drag the chart (click and drag)

3.6 TO VIEW YOUR WORK BEFORE PRINTING

- i. Click on the file menu
- ii. Click print preview
- iii. When done click close

But within the print preview you can adjust your settings if you so wish to.

To print

- i. Select the range of data to be printed
- ii. From the file menu click print
- iii. Select the printer option
- iv. Select the print range

- v. Select the number of copies to print
- vi. Click on prints

4.0 COREL DRAW

This is a window based graphic application that is used in organizing text and graphic electronically.

4.1 THINGS TO NOTE

- i. **Drawing Page:** This is rectangle shape which indicates a working space for all your designs. It is object that appears on the drawing page that would be printed.
- ii. **Colour palette:** This contain colours that can be applied on outline information or to fill an object.
- iii. **Status Bar:** This display information about the ongoing operation in the computer.
- iv. **Tool Bar:** This give quick access to available tools, some tools have fly out (it helps to create, fill and modify your drawing).

4.2 USES OF COREL DRAW

- i. For complex drawing
- ii. Fill colour illustration
- iii. Logo
- iv. Cover page (for newspaper, magazines etc)
- v. Cards (seasons, wedding etc)
- vi. Certificates etc

4.3 LOADING COREL DRAW

- i. Click on the start button
- ii. Click on “programs”
- iii. Click on “corel graphic suite”
- iv. Click on “corel draw” (5,6,7,8,9,10,11,12,13) or

Double click the “corel draw icon” on your desk top.

Once you click corel draw the follow option will appear on your screen.

- New graphic
- Open last edited
- Open graphic
- Template
- What's new

4.4 FOR A NEW DRAWING

- i. Click on new graphic icon or click file
- ii. Click new

A rectangular shape space will appear where you can use your tools to start drawing.

4.4.1 TO OPEN AND EXISTING DRAWING

- i. Click on file menu
- ii. Click open
- iii. Select your drive
- iv. Select the file name
- v. Click open

4.4.2 TOOLS USED

Pick Tool: This is an arrow like shaped icon used to select, resize, rotate skew and move object from one location to another, shape tool, eclipse tool.

To use the pick tools follow the steps;

- i. Double click the pick tool to select object or multiple object
- ii. Click on one of the object
- iii. Hold down the shift key
- iv. Click on other objects one after the other

4.4.3 RESEIZING AN OBJECT

- i. Click on the pick tool
- ii. Click on the object (about 8 tiny nodes appear around it)

- iii. Place the cursor on one of the nodes and drag to increase or reduce the size

4.4.4 ROTATING AN OBJECT

- i. Click on the pick tool
- ii. Double click the object (some arrow appear around it)
- iii. Place your cursor on one of the arrow and rotate

4.4.5. MOVE AN OBJECT

The drag and drop

- i. Select the object with the pick tool
- ii. Drag the object to new location or
Use arrow key
- i. Select the object with the pick tool
- ii. Control the object to new location (using the arrow key)

4.4.6 EDITING

To edit, select the object make correction

4.4.7 TEXT

To write text with the pick tool.

- i. Click on the text tool
- ii. With the text tool on the following page drag to create a box
- iii. Inside the box, type your text

The are two (2) types of text

- a. **Artistic text:** Used for short form of text that can have effects applied to them

- b. **Paragraph text:** Use for longer blocks of text can be extensive by formatted.

4.4.8 TO ALIGN TEXT

- i. Select the text
- ii. Click text, from dialogue box
- iii. Click align

- iv. Select left, right or centre

4.4.9 TO ADJUST FONT SIZE, STYLE, ETC

- i. Select the text
- ii. Click text, from the dialogue box
- iii. Click format
- iv. From the box click what you want to do.

4.5 SAVING A DRAWING

- i. Click file
- ii. Click “save as” (to save for the first time)
- iii. Click save (if not for the first time)
- iv. Choose drive of your choice
- v. Type in file name in the box
- vi. Click save or
- vii. Click on the close button on top of screen (x)

4.5.1 CLOSING A FILE

- i. Click file menu
- ii. Click close or
- iii. Click on the close button on top of screen (x)

4.5.2 TO CLOSE AN APPLICATION

- i. Click on file menu
- ii. Click exit

4.5.3 TO SPECIFY THE PAGE LAYOUT

- i. Click layout on the menu bar
- ii. Select page set-up
- iii. Choose portrait or landscape (for page layout) A3, A4 etc for paper size.
- iv. Click of

4.5.4 TO SET RULER

- i. Right click the ruler

- ii. Choose ruler set-up
- iii. Select cm or inches etc

4.5.5 IMPORTING OBJECT

- i. Click on the file menu
- ii. Click import
- iii. Open source from the look in
- iv. Double click on the dragging page

4.5.6 EXPORTING OBJECT

- i. Select object
- ii. Click on file menu
- iii. Click export
- iv. Click destination
- v. Elect file name
- vi. Click export

4.5.7 DELETE AN OBJECT

- i. Select the object using the pick tool
- ii. Click edit menu
- iii. Click delete or
- iv. Select the object
- v. Press the delete key

4.5.8 TO PERVERSE YOUR PREVIOUS ACTION

- i. Click edit menu
- ii. Click undo or
- iii. Press Ctrl+Z on the keyboard

4.5.9 GROUPING OBJECT

- i. Select all with the pick tool
- ii. Click arrange menu
- iii. Click group

- iv. Double click the pick tool and press ctrl+C key on the key board.

4.5.10 UNGROUPING OBJECT

- i. Select the grouped object
- ii. Click arrange menu
- iii. Click ungroup or
- iv. Select the grouped object
- v. Right click on the selected object
- vi. Click ungroup

Undo, enable one reverse his previous action

4.6 CHANGING AND ORDER OF OBJECT

When you have multiple objects overlapping one another in your drawing, this command is used to rearrange them according.

- i. Select the object to be rearrange
- ii. Click arrange menu
- iii. Select order and choose either to form or back etc.

4.7 SHAPE TOOL

This tool is used in rounding corners of object e.g rectangle number and point of polygon or star etc.

4.7.1 ROUND THE ANGLE OF A RECTANGLE/SQUARE

- i. Select the rectangle square using shape tool
- ii. Drag one of the nodes inwardly to round

4.7.2 TO SHAPE A CIRCLE OR ELLIPSE INTO AN ARC

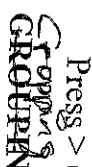
- i. Select the object with the shape tool
- ii. Drag one of the nodes inwardly
- iii. Drag with the cursor inside the perimeter

4.7.3 KERNING CHARACTER

Kerning means increasing or decreasing the space between characters, words or line.

- i. Click text menu

- ii. Click the spacing
- iii. Select inter-character, interword or inter-line spacing or
- iv. Select text with the pick tool
- v. Press and hold ctrl + shift keys

vi. Press > (greater than) sign non the keyboard to increase between characters.
 **CROPPING BITMAP**

Cropping a bitmap means to cut unwanted parts of the bitmap (a graphical image) it reduces visible area and he original size.

- i. Open the shape edit fly out and select the shape tool
- ii. Select the bitmap with the shape tool
- iii. Drag the bitmap node to change the shape of the bitmap outlines
- iv. Click on the pick tool
- v. Click the bitmap and crop the bitmap

4.8.1 ERASER TOOL

This tool let you remove portions or part of an object completely. To erase portion of an object;

- i. Select the pick tool
- ii. Select the object with the pick tool
- iii. Open the shape edit fly out and select the eraser tool
- iv. Drag over the object

4.8.2 RECTANGLE TOOL

It helps to draw rectangles and squares.

- i. Select the rectangle tools
- ii. Position the cursor where you want the rectangle to appear
- iii. Drag diagonally to draw the rectangle for square.
- iv. Repeat steps 1 & 2 above
- v. Hold down ctrl key and drag diagonally

4.8.3 GRAPH PAPER

To draw a graph paper;

- i. Select the graph paper tool in the tool box
- ii. Take the cursor where you want it to appear
- iii. Drag down diagonally to draw the graph paper to your desired size.

4.8.4 ECLIPSE TOOL

It lets you draw eclipse and perfect circles.

- i. Click to select the eclipse tool in the tool box
- ii. Place the cursor where you want the eclipse to appear
- iii. Drag down diagonally

For perfect circle;

- i. Click the eclipse tools in the toll box
- ii. Place the cursor where you want the circle to appear
- iii. Hold down the ctrl key on the keyboard
- iv. Drag down diagonally

4.8.5 TO HIGHLIGHT A TEXT

- i. click on the text to be highlighted with the pick tool
- ii. it click the text tool on the tool box or
- iii. double click on the text and drag to the length you want with your cursor or
- iv. left click on the text and drag your cursor to the length you want or
- v. press ctrl+A on the keyboard to highlight all

4.8.6 INTERACTIVE TRANSPARENCY TOOL

This allow you to add colour to your drawing

4.8.7 TO APPLY FONTAIN

- i. Click to select the object with the pick too
- ii. Select the interactive fill tool and select a colour in the colour pollette

4.9 INTERNET

The word Internet simple stands for international network

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lets
mb
78

5.1 DEFINITION OF SOME ACRONYM

- i. **Network of Computers:** This is a two or more computers connected together using cable or other connection means to communicate.
- ii. **Internet:** It is a collection of computer around the globe that are connected to each other by telephone lines and satellite line.
- iii. **Internet Bandwidth:** This is a telecommunication ward that describe how much data can be transmitted per second over the internet. It is measured in bits per send (bps) kilobi per second (kps) megabits per send (mps) etc. The amount of bandwidth you have available through your internet service provided will determine how quickly you will be able to access or download data on the net.
- iv. **World wide web (www):** This is the multimedia part of the internet, it is a sub-set of the internet that provide a friendly way of organizing and viewing a large portion of the information available on the internet it's content is displayed as a page, text, formatted in various styles and categories which can contain pictures ^{N, photos} ~~video~~ etc. The initial screen you see when you access a website is usually referred to as a Home Page
- v. **Web Browser:** A client program that is used to display the information provided by a specific server to which you have connected it gives you an opportunity to view browser for information on the internet e.g Microsoft internet explorer etc
- vi. **Hypertext Transfer Protocol (HTTP)^(S):** A standard language that allows communication between web clients and server. Hypertext is used to describe an interlinked system of documents in which a user may jump from one document to another through the help of hyper links (portion of hypertext document that are linked to other related documents on the net.

5.2 APPLICATION INTERNET

Internet is applicable in so many areas such as.

- i. Electronic mail (e-mail)
- ii. Electronic Banking (e-banking)
- iii. Electronic commerce
- iv. Electronic learning (e-learning)
- v. Electronic library (e-library)
- vi. Electronic news (e-news)

Write

Electronic Mail (e-Mail): It is used to send ~~write~~ message between individuals or group often geographically separated by large distance. It is a very convenient but expensive way to transmit message, it has become a key part of the communication network of most modern office and business organization.

Electronic Banking (e-banking): The Nigerian banking sector has witnessed a greater change and rapid development which manifested in the enormous volume and complexity of operations, increased innovation and variety in product/services rendered financial liberalization and business process also included such as:

- i. Payment facilities by credit and
- ii. Account verification
- iii. Online access to bank accounts

Electronic learning (e-learning): With the help of the internet, people are able to study online wherever one might find himself at a particular time on-line easier because you musn't be there in person or proxy to achieve your aim as far as there is mobile network.

Electronic commerce (e-commerce): This is a business process the involve the use of the internet as a means of transaction over the globe using the global of area network (GAN) or the local area network (LAN), such business include; online transaction payments, electronic payment of transaction, electronic shopping etc.

Electronic Library (e-library): Availability of the internet makes it possible for one to have access to many library facilities on-line with these you don't need to go to the

library before having access to books, journals, news papers, research materials etc. it the mobile network you can access all these needs on line.

Electronic News (e-News): This refers to online news, which different news papers and news agencies such as (ww.BBc, Radio finance, Vanguard news paper, Punch news paper, etc can all be access via the internet with the use of mobile network.

5.3 HOW TO SEND AND RECEIVE E-MAIL

Things to ~~not~~^{do} before sending and receiving an e-mail

- a. Sign-Up: This process involves the creating of an e-mail account or box to enable you get or user identity (user ID) and password a new account e.g on yahoo.mail, Hotmail, gmail, etc follow the following steps:
 - i. Type in your yahooemail address i.e www.yahooemail.com or for Hotmail type www.msn.com a yahoo home page will appear.
 - ii. Click on sign-up: Another page will appear
 - iii. Supply all the information requested on you
 - iv. Click on the submit this form button is there is error in the information you supplied it will be displayed for you to effect correction and resubmit.

- b. Your e-mail address now look like the Jommi@yahoo.com which comprises of Jommi (person name)

Yahoo (domain name)

Com (domain type)

5.3.1 OPENING/SIGN-IN YOUR MAIL BOX

Sign in or logging in is the process of opening an existing e-mail box using the user ID and password” as an access key.

To sign-in or login one must;

- i. First open the webpage of the e-mail service by typing the web address in the address bar.
- ii. Enter the user name or ID in the text bar provided
- iii. Enter the appropriate password of the ID

- iv. Click on the sign in/login button or pulss enter key is you are using a computer

5.3.2 COMPOSING/SENDING A NEW MESSAGE

To compose a new message to send to someone

- i. Click the compose/new email button in the tool bar
- ii. The window will open up.
- iii. Click on the address box and type the recipient address
- iv. To send message a others, click in the "cc" text box and type their e-mail address separating each with a comma, you don't want the receiptence to see the address of each other then click the "BSC" instead of "CC"
- v. Click on the subject box and type a short description of what your message is about.
- vi. Click in the message window and type your mail.

5.3.3 RECEIVING AND READING MESSAGE/MAIL

To receive and read your mail;

- i. Click on the inbox or check mail to display your mail, your inbox shows the number of mail in the inbox, date received, sending's address and time received.
- ii. To open and read the mail, click on the mail subject and the mal will be ready to used.
- iii. To delete any unwanted mail, click the box by the left headside you want to delete and click on delete button.
- iv. To replay the sender click on the reply button type your reply and click send.
- v. To forward a mail to someone click on the forward to button and type the recipient address.

5.3.4 ATTACHING FILES

To attach a file or files;

- i. Click on the attach file link
- ii. Click on browse link
- iii. Using look in drop down list and check the folder that contain the file you want to attach and click open after selecting the file.
- iv. Click on done button

5.3.5 SIGN-OUT/LOG-OUT

After you must have finished all your needs in your mail box next is to sign out.

- i. Close any open mail box
- ii. Click on the sign out/log out button.

5.4 SAVING INTO STORAGE DEVICES

To save the webpage information into storage devices such as flash, hard disk etc

- i. Open the page you want to save
- ii. On the menu bar click file
- iii. From the pull down menu, click on "Save as"
- iv. Select where you want to save the document from the dialogue box, to save into a floppy disc insert the floppy disc into the disc drive and click save.
- v. To save to hard disc select the drive C and click save.
- vi. If the page contain images, right click on it choose save picture as the save.

5.5 PRINTING OUT MATERIALS FROM WEB PAGE

When you want to print document directly

- i. Click on edit on the menu bar
- ii. Click on select all
- iii. Click on the file on the menu bar
- iv. Click on print a print dialogue box appear
- v. Click Ok

5.6 TO DOWNLOAD SOFTWARE FROM THE NET

When you want to download any software from the net, make sure that it is from a xirus free site. After locating the software do the following.

- i. Click the pages, hyperlink to begin
- ii. Once the process is activated a file download dialogue box will be displayed, this is always a safety strategy to protect this is system against software virus.

From the dialogue box select any option

- a. Run this program from its current location
- b. Save it to disc
- c. Then click the Ok button
- iii. While the download is in process, a message box keeps you informed of the file download process.
- iv. Click cancel button to stop the download when necessary

5.7 HANDLING ERROR MESSAGES

When cheating or browsing you may encounter the error message such as page cannot be found, incorrect URL etc when such happen try.

- i. Connecting again or click refresh to update
- ii. Click the back button to go back to previous page
- iii. Check for possible error in the webpage address or URL address
- iv. If no problem found or cannot connect again check your internet connection settings.

5.8 PROBLEMS OF INTERNET USAGE IN NIGERIA

- i. Poverty level
- ii. Shortage of professionals and skilled manpower
- iii. Lack of government support
- iv. In adequate power supply
- v. Low literacy level, poor knowledge on the internet usage
- vi. Piracy and insecurity to sensitive information
- vii. Ineffectiveness of the regulatory bodies